Imitation Learning

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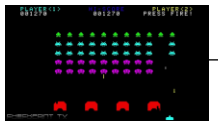


RL

a_1 ?

$\boldsymbol{a_2}$ $\mathbf{0.3}$

a_3 ?

IL

a_1 0

$\boldsymbol{a_2}$ $\mathbf{1}$ (expert choice)

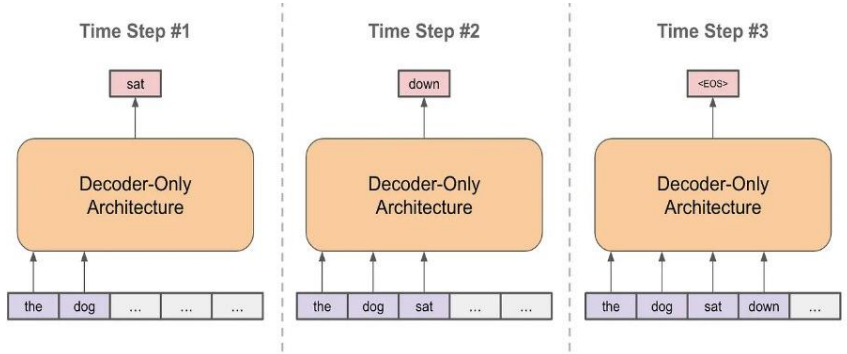
a_3 0

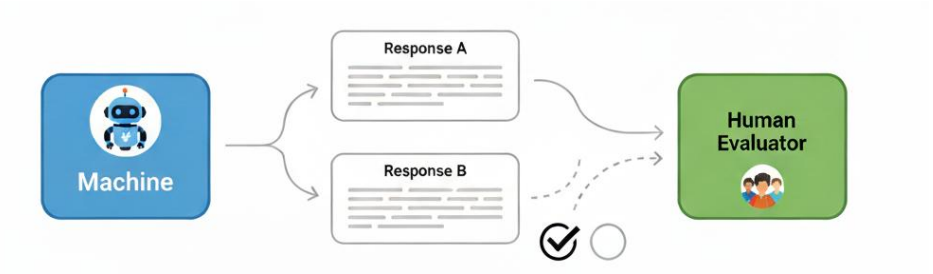
Offline IL: learn from static data generated by the expert $\{(s_1, a_1^\star, s_2, a_2^\star, \dots, s_H, a_H^\star)\}$

Online IL: may interact with MDP and query the expert $\{(s_1, a_1, a_1^\star, s_2, a_2, a_2^\star, \dots, s_H, a_H, a_H^\star)\}$

Examples

- Language models





- Robotics

Types

- Direct Imitation: directly learn policy to imitate the expert
  - Behavior cloning
  - DAgger
  - Direct preference optimization (preference feedback)
- Inverse RL: learn reward function from expert, and perform RL on it
  - Adversarial IRL (paper)
  - MaxEnt IRL (paper)
  - RLHF (preference feedback)

Direct Imitation

Behavior Cloning: Reduction to Supervised Learning

Behavior Cloning (Offline IL)

Run expert policy $\pi^\star$ and obtain $(s_1, a_1), \ldots, (s_N, a_N)$

Train a policy π_θ such that

$$\pi_\theta(s_i) \approx a_i$$

(classification if actions are discrete)
(regression if actions are continuous)

Behavior Cloning: Reduction to Supervised Learning

Behavior Cloning (Offline IL)

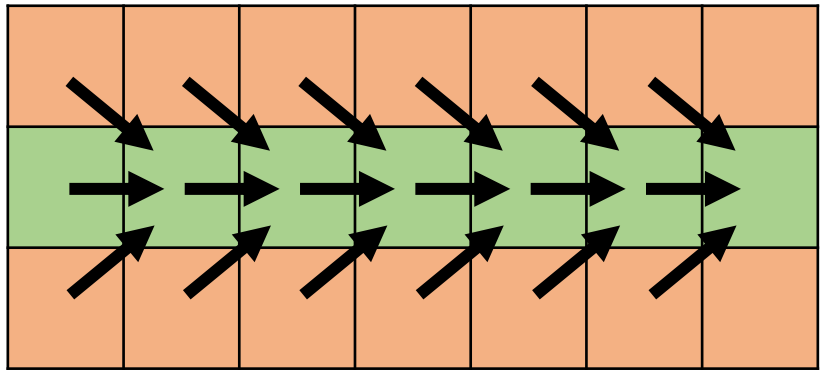
Run expert policy $\pi^\star$ and obtain $(s_1, a_1), \ldots, (s_N, a_N)$

Obtain policy π_θ by finding

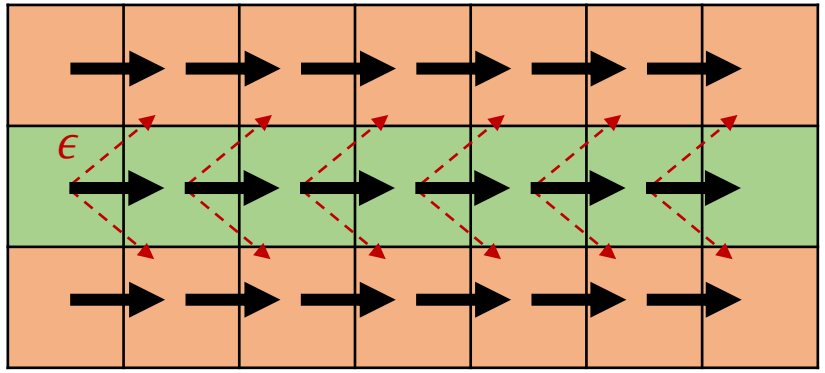
$$\theta = \underset{\theta}{\operatorname{argmin}} \ \frac{1}{N} \sum_{i=1}^{N} \log \frac{1}{\pi_\theta(a_i | s_i)} \qquad \text{(discrete action)}$$

$$\theta = \underset{\theta}{\operatorname{argmin}} \ \frac{1}{N} \sum_{i=1}^{N} \|a_i - \pi_\theta(s_i)\|^2 \qquad \text{(continuous action)}$$

Behavior Cloning: Reduction to Supervised Learning



Optimal policy

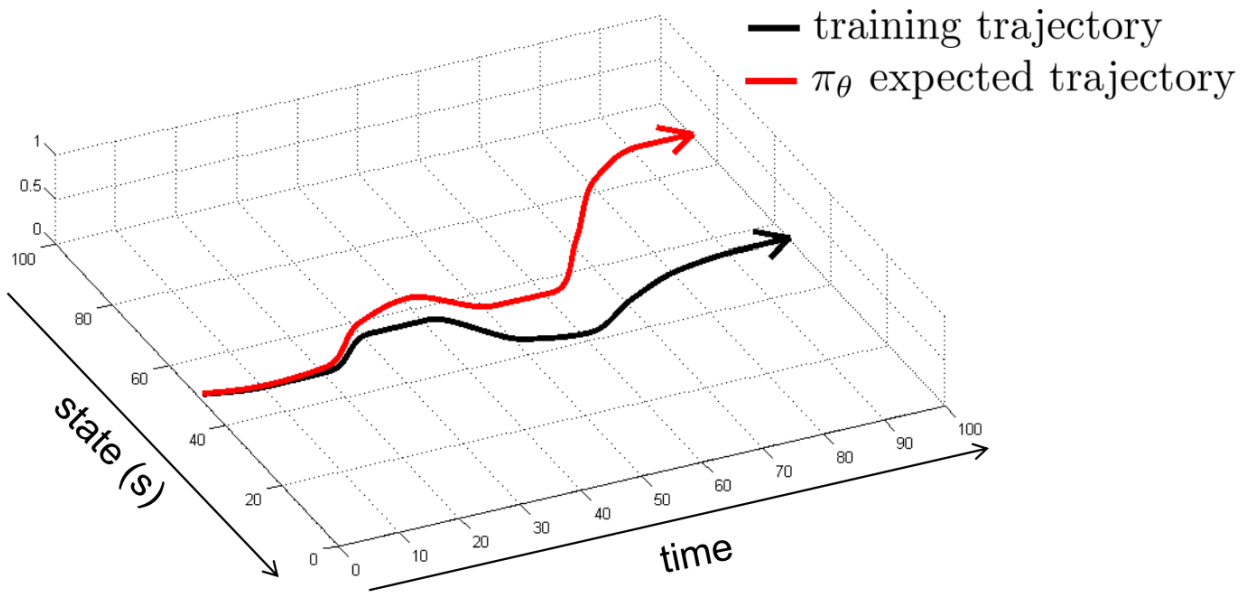


A policy with low training error (ϵ)

but high testing error ($\approx \epsilon \times$ Horizon_length)

Key issue: the training data does not cover the states visited by the learner's policy

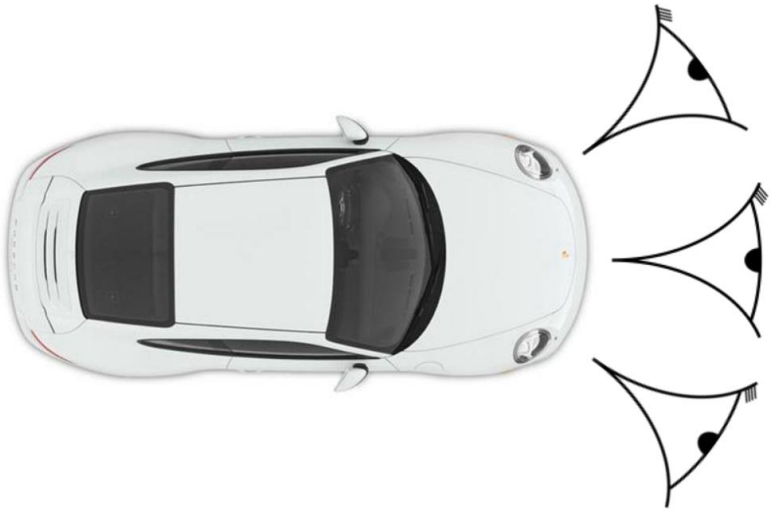
Behavior Cloning: Reduction to Supervised Learning



Distribution shift

Solution 1

- Data augmentation



Bojarsky et al. End to End Learning for Self-Driving Cars. 2016

https://www.youtube.com/watch?v=NJU9ULQUwng

Solution 2: Interact with Expert (Online IL)

Dataset Aggregation (DAgger)

For $k = 1,\ 2, \ldots$

Train $\pi_\theta(a|s)$ with dataset $\mathcal{B}$

Run π_θ to collect states $s_1, s_2, \ldots, s_N$

Ask the expert to label actions, giving $(s_1, a_1), \ldots, (s_N, a_N)$

Add $(s_1, a_1), \ldots, (s_N, a_N)$ to $\mathcal{B}$

Stephane Ross, Geoffrey J. Gordon, J. Andrew Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

Solution 2: Interact with Expert (Online IL)

A Variant of DAgger

For $k = 1,\ 2, \ldots$

Train $\pi_\theta(a|s)$ with dataset $\mathcal{B}$

Run π_θ

Ask the expert to take over at some time step

Store in $\mathcal{B}$ the samples generated by expert intervention

Stephane Ross, Geoffrey J. Gordon, J. Andrew Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

Key: collecting (s, a) samples with s generated from the learner (+ expert), and a from the expert

Imitation with Preference Feedback

Direct Preference Optimization (DPO) (link) (An important technique for LLM alignment)

Given context x and two potential actions, the expert chooses the **preferred** one (the preferred one is denoted as y_w and the other is y_l).

$$\mathcal{L}_{\mathrm{DPO}}(\pi_\theta; \pi_{\mathrm{ref}}) = -\mathbb{E}_{(x, y_w, y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\mathrm{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\mathrm{ref}}(y_l \mid x)}\right)\right]$$

π_{ref} is a reference policy that π_θ is initialized as.

The previously discussed frameworks of online / offline IL can be directly applied here.

Inverse RL

Inverse Reinforcement Learning (IRL)

Find a reward / value function that explains the behavior of the expert.

- **Behavior assumption:** Assume some connections between expert policy and reward/value. E.g.,

  $\pi^\star(a|s) \propto \exp(Q^\star(s,a))$

  $P^{\pi^\star}(\tau) \propto \exp\left(\sum_h R^\star(s_h, a_h)\right)$

- **Maximum likelihood:** Given expert trajectories, find ϕ that maximizes likelihood
- **Policy training:** Find a policy that approximately maximizes the expected reward / value under ϕ. E.g.,

  $\pi(a|s) \propto \exp\left(Q_\phi(s,a)\right)$

  $\theta = \underset{\theta}{\operatorname{argmax}}\ \mathbb{E}_{s\sim D}\left[\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[R_\phi(s,a)\right] - \beta\ \mathrm{KL}(\pi_\theta(\cdot\,|s), \pi_{\mathrm{ref}}(\cdot\,|s))\right]$

Mathematically, this "learn reward from expert + learn policy from reward" paradigm can always be condensed as "learn policy from expert"

IRL with Preference Feedback

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp\left(r^*(x, y_1)\right)}{\exp\left(r^*(x, y_1)\right) + \exp\left(r^*(x, y_2)\right)}$$

Behavior assumption

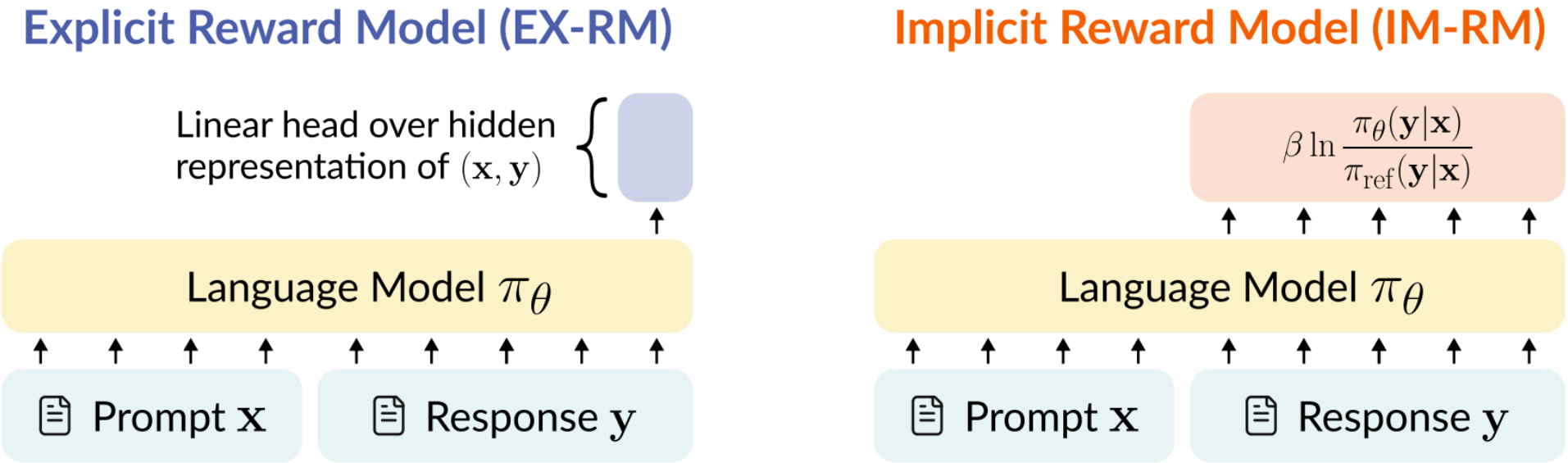
$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))\right]$$

Maximum likelihood

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(y|x)}\left[r_\phi(x, y)\right] - \beta \mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(y \mid x) \mid\mid \pi_{\mathrm{ref}}(y \mid x)\right]$$

Policy training

Benefits of having an explicit reward model



Razin et al., 2025. Why is Your Language Model a Poor Implicit Reward Model?